



## Jahid Hasan Rony

Research Assistant, RIoT Research Centre, IUB

Email: [research.rony@gmail.com](mailto:research.rony@gmail.com)

Cell: +8801753296896

Google Scholar: <https://scholar.google.com/citations?user=3ScF50oAAAAJ>

Web: <https://rony-cse.github.io/>

Linkedin: <https://www.linkedin.com/in/j-h-r>

### CAREER GOAL

#### Teaching Statement

Throughout my experience as a teaching assistant at Independent University Bangladesh, I have had the opportunity to teach courses such as "Fundamental of Computer Systems" and "Programming in Python." My teaching philosophy is centered around clarity, hands-on engagement, and the cultivation of curiosity among students. I believe that effective teaching in Computer Science & Engineering requires an interactive and incremental approach, where each concept builds upon the previous one, allowing students to see their progress clearly. In my teaching, I emphasize the importance of blending theoretical knowledge with practical applications, ensuring that students not only understand the concepts but also see their relevance in real-world scenarios.

#### Research Statement

My research focuses on applying machine learning, deep learning, and real-world challenges, with a particular emphasis on UAV (drone) technology. I have basic knowledge on ML-based UAV applications, such as real-time obstacle detection and environmental monitoring, bridging the gap between theoretical research and practical implementation. I am looking forward to contribute innovative methodologies that enhance the capabilities, pushing the boundaries of my knowledge in various industries while advancing academic knowledge.

### EDUCATION

Nov, 2018 – Aug, 2023      **B.Sc.** Computer Science and Engineering

Dhaka University of Engineering and Technology

Jan, 2012 – Sep, 2016      **Diploma** in Computer Technology

Bogura Polytechnic Institute. Bangladesh.

### SELECTED PUBLICATIONS

- [C.3]  
2024      **Jahid Hasan Rony**, M. M. Mahbubul Syeed, Razib Hayat Khan, Kaniz Fatema, Md Shakhawat Hossain, Mohammad Faisal Uddin, "Predicting Depression among University Students: a Comparative Assessment of ML & DL models using XAI," *23rd International Symposium on Communications and Information Technologies (ISCIT) 2024*, Thailand [**Rank: B**, Scopus Indexed, IEEE]
- [J.6]  
2024      S. Rahman, **J. H. Rony**, J. Uddin, "Real-time Obstacle Detection with YOLOv8 in a WSN Based on UAV Aerial Photography," *J. Imaging | Journal of Imaging | ISSN: 2313-433X* [**Rank: Q2**]
- [J.5]  
2023      M.K. Kabir, A.N.B. Kabir, **J. H. Rony**, J. Uddin, "Drone Detection from Live Stream Video Using Image Processing Techniques and YOLOv7, *I.J. Image, Graphics and Signal Processing*, 2023, 4, 1-4 DOI: 10.5815/ijigsp.2020.04.01 [**Scopus Indexed**]
- [J.4]  
2021      **J. H. Rony**, N. Karim, M. A. Rouf, M. M. Islam, J. Uddin, and M. Begum, "A Cost-Effective IoT Model for a Smart Sewerage Management System Using Sensors," *MDPI J*, vol. 4, no. 3, Art. no. 3, Sep. 2021, doi: 10.3390/j4030027
- [C.2]  
2024      **J. H. Rony**, N. Karim, M. A. Rouf, and F. H. Siddiqui, "Mosquito Species Classification through Wingbeat Analysis: A Hybrid Machine Learning Approach," *International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM 2023)*, Gazipur, Bangladesh: IEEE, Jun. 2023
- [C.1]  
2023      Razib Hayat Khan, Jahid **Hasan Rony**, Mahbubul Syeed, Performance Analysis of Food Delivery Service Using UML and SPN, *23rd International Symposium on Communications and Information Technologies (ISCIT) 2024*, Thailand [**Rank: B**, Scopus Indexed, IEEE]
- [J.3]  
2021      M. Z. Hossain, M. J. Islam, M. W. R. Miah, **J. H. Rony**, and M. Begum, "Develop a dynamic DBSCAN algorithm for solving initial parameter selection problem of the DBSCAN algorithm," *Indonesian Journal of Electrical Engineering and Computer Science*, 2021, doi: 10.11591/ijeecs.v23.i3.pp1602-1610 [**Rank: Q3**]

- [J.2] 2023 Md. M. Islam, **J. H. Rony**, Md. N. Akhtar, S. U. P. Shakil, and J. Uddin, "Water Monitoring Using Internet of Things," in EAI/Springer Innovations in Communication and Computing. Cham: Springer, 2023, 10.1007/978-3-031-09729-4 [Rank: Q4]
- [J.1] 2023 M. Begum, **J. H. Rony**, M. R. Islam, and J. Uddin, "A Long-term Software Fault Prediction Model Using Box-Cox and Linear Regression," J. Inf. Syst. Telecom. JIST, vol. 3, no. 43, p. 222, Aug. 2023. [Rank: Q4]

## EMPLOYMENT [ 1 Years of Professional Experience ]

**15 Oct, 2023– Ongoing**

**Teaching Assistant.**

Dept. of Computer Science and Engineering. Independent University, Bangladesh (IUB).

**15 Oct, 2023– Ongoing**

**Research Assistant.**

Coding for All Initiative, VC office, Independent University, Bangladesh (IUB).

## AWARDS AND ACHIEVEMENTS

|                           |                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| International Conference  | Best Poster Presentation at International Conference on Sustainable Engineering Development (ICSED - 2023). |
| International Competition | Champion, Techfest IIT Bombay 2023, Mesmerize Segment                                                       |
|                           | Top-10, World Robotics Championship in India, 2023                                                          |
|                           | Champion, IEEE R10 Quiz Competition, arranged by IEEE Hyderabad Section                                     |
|                           | Champion at Mind Spark 2022, Arranged by AUST                                                               |
| National Competition      | Champion at DRC Fest 2022, Arranged by DUET                                                                 |
|                           | Champion, Robotics Competition at NACTAR 2022                                                               |
|                           | Placed Among Top-30, Mujib-100 idea competition, Arranged by UGC                                            |

## PROJECT DEVELOPMENT

### IoT Based Real Time Sewerage Monitoring System

- Integrating microcontrollers and sensors to collect real-time data, stored in a cloud database
- Developed a web application using with Flutter over Firebase and Thingspeak

### Developing and maintaining DUET Robotics Club Official Website

- Developed and maintained the new official website of DUET Robotics Club since 2020

### Mobile Apps & IoT Based Transformer Monitoring & Safety Mechanism

- Built a cost-effective IoT device for monitoring electrical transformer load and implementing safety protocols, seamlessly integrating real-time data to mobile app.
- This project highlights my skills in IoT, mobile app development and automation.

## TECHNICAL SKILLS

|                       |                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Language              | C/C++, Python, Matlab.                                                                                                                            |
| Platform              | PyCharm, Visual Studio Code, Colab, Kaggle                                                                                                        |
| Framework             | TensorFlow, Keras. Library: Librosa, Scikit-learn, NumPy. Data Manipulation and Analysis: Pandas, Seaborn, Matplotlib, wordPress, Flutter, EEGLAB |
| Developer Tools       | Git, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse                                                                                           |
| Embedded System       | Arduino, ESP, Raspberry PI   Design Software: Proteus, Blender   Data Management: Firebase, ThingSpeak.                                           |
| Image & Video Editing | Adobe Photoshop, Illustrator, Filmora                                                                                                             |

## RESEARCH INTEREST- KEYWORDS

Machine Learning, Deep Learning, Computer Vision, UAV.

## LEADERSHIP/ CO-CURRICULAR ACTIVITIES

**DUET Robotics Club:** President (2022-2023), General Secretary (2021-2022)

**National Robotics Festival:** Convener of DUET Techfest 2023, DRC Fest 2022, Co-convener DUET Techfest 2022

**International Robotics Competition:** Head of organizing committee of Techfest IIT Bombay Bangladesh Round 2023

**International Conference:** Volunteer of NCIM 2023 (arranged by dept of CSE, DUET) and ICAEEE 2022 (dept of EEE)

## PERSONAL INFORMATION

**Language** English (professional), Bengali (mother tongue)

**Date of Birth** 26-08-1997

**Nationality** Bangladesh.

**Marital Status** Single

## REFERENCE

### MM Mahbubul Syeed, PhD

Associate Professor, Department of Computer Science and Engineering,  
Independent University, Bangladesh

Email: [mahbubul.syeed@iub.edu.bd](mailto:mahbubul.syeed@iub.edu.bd)

### Mohammad Faisal Uddin, PhD

Associate Professor, Department of Computer Science and Engineering  
Independent University, Bangladesh

Email: [faisal@iub.edu.bd](mailto:faisal@iub.edu.bd)

### Fazlul Hasan Siddiqui, PhD

Professor, Department of Computer Science,  
Dhaka University of Engineering and Technology.

Phone: +8801957964549

Email: [siddiqui@duet.ac.bd](mailto:siddiqui@duet.ac.bd)

### Dr. Momotaz Begum

Professor, Department of Computer Science,  
Dhaka University of Engineering and Technology.

Email: [drmomotaz@duet.ac.bd](mailto:drmomotaz@duet.ac.bd)